HAFIDZ MUTA'ALI

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I am a Machine Learning Engineer specializing in Computer Vision and AI system deployment, with experience developing and productionizing deep learning solutions. My work covers end-to-end AI pipelines including object detection, segmentation, classification, and OCR systems. I have expertise in scalable AI infrastructure, optimizing inference pipelines, and deploying GPU-based solutions using Docker, TensorRT, and distributed frameworks. I work with high-throughput video analytics, multi-stream inference, and NVIDIA GPU optimization, and have deployed edge AI solutions on edge devices integrated with Azure IoT Hub for reliable cloud-edge video analytics.

Work Experiences

Sigmawave AI - **Singapore (Remote)** Nov 2025 - Present

Computer Vision Engineer Lead

- Led the development and deployment of YOLO-based models for firearm detection, ensuring strict alignment with MINDEF security and compliance requirements.
- Spearheaded the transition to Oriented Bounding Box (OBB) annotations, significantly improving localization precision for rotated and partially occluded objects compared to standard axis-aligned methods.
- Architected synthetic data generation pipelines to augment real-world datasets, drastically reducing false negatives in edge cases involving motion blur, high-velocity objects, and extreme camera angles.
- Mentored a team of three engineers on model architecture selection, training strategies, and evaluation metrics, fostering a culture of technical excellence and deployment readiness.
- Engineered high-performance inference pipelines for both NVIDIA and AMD ecosystems, achieving an 11x speedup (380 FPS) on AMD Instinct MI325X (ROCm/MIGraphX) and a 3x speedup (65 FPS) on RTX 4080 Super (TensorRT) for 4 concurrent video streams.
- Optimized throughput via strategic batch-32 processing and graph-level optimizations (operator fusion and precision calibration), maximizing GPU utilization for real-time firearm detection.

PT. Freeport Indonesia - **Jakarta, Indonesia** May 2023 - Present

Machine Learning Engineer

- Optimized a scene detection model by expanding its classification capabilities from two categories (material and background) to three categories (water, material, and background). The enhanced model achieved an overall accuracy of 91.1% with the following performance metrics
- Deployed solutions using Snowflake Python Procedure, reducing data latency and ensuring data remains within the Snowflake environment. This approach significantly minimized the risk of data leakage, enhanced data security, and improved real-time data processing speed, ultimately accelerating development and deployment time to production.
- Developed a deployment pipeline using IoT Hub to manage multiple devices. When a new model update is available, the updated Docker image is stored in Azure Container Registry (ACR) and automatically deployed to each device via IoT Hub.
- Optimized Docker images by reducing size and improving build efficiency. Streamlined dependencies, Ensured consistency by avoiding the latest tag and pinning specific versions, while also optimizing layer caching by properly ordering RUN, COPY, and ADD commands
- Developed and implemented scheduling mechanisms using RMJ for various tasks and processes, enhancing automation and ensuring timely execution of critical operations. This led to increase in task efficiency and reduced manual oversight requirements.
- Conducted script refactoring initiatives to improve code quality, readability, and maintainability. This resulted in more robust and scalable solutions for ongoing projects.
- Collaborated with cross-functional teams to identify and address challenges in data science and machine learning projects.

PT. Inti Sinergi Teknologi - **Jakarta, Indonesia** Feb 2022 - May 2023

Machine Learning Engineer

- Optimized bounding box reading in a PyTorch function within PARSeq for text detection, ensuring that bounding boxes do not get cut off or overlap with other text.
- Implemented automated solutions for the extraction of data from customer purchase order PDF documents. This involved dealing with both scanned and system-generated documents by creating and applying templates. Utilized OCR technology to identify and retrieve specific fields, streamlining the data extraction workflow.
- Spearheaded research and development initiatives focused on machine learning techniques to extract textual content and tables from PDF files. Developed algorithms with the capability to export the extracted data into CSV format, ensuring compatibility with various data analysis tools.
- Creating comprehensive datasets and developing algorithms for training machine learning models. These models were specifically designed to extract targeted information from PDF documents, enhancing the accuracy and efficiency of data extraction processes.
- Conducted research on named entity recognition techniques and successfully implemented a transformer model for this purpose. The NER system was designed to identify and classify entities, contributing to improved understanding and categorization of textual data.

Education Level

Universitas Pembangunan Nasional Veteran Yogyakarta - **Yogyakarta, Indonesia**
Bachelor of Informatics, 3.46/4.00

Hacktiv8 - **Jakarta, Indonesia**
Certificate in Full Stack Data Science, 87.48/100.00

- Developed proficiency in image classification techniques, learning to train models to accurately categorize images into predefined classes.
- Gained practical experience in object detection, utilizing frameworks like TensorFlow or PyTorch to identify and locate objects within images.
- Acquired skills in plate number detection, a critical aspect of computer vision applications, especially in the context of traffic monitoring and security.
- Implemented algorithms and models capable of detecting license plates in images, followed by recognition of alphanumeric characters.
- Explored and implemented drowsiness detection systems using computer vision and machine learning.

Skills

- **Programming and Libraries**: Proficient in Python programming language. Expertise in using essential libraries such as Pandas, Numpy, and Matplotlib for data manipulation and visualization. Skilled in Scikit-Learn for machine learning applications. Experience with deep learning frameworks like TensorFlow and PyTorch.
- **Tools and Platforms**: Experience in using SQL for database querying and manipulation. Experience with Google Colaboratory and Anaconda Navigator for collaborative and interactive coding. Experience with Azure DevOps and Azure ML for managing and deploying machine learning models. Competent in using Visual Studio Code as an integrated development environment (IDE). Experience with Docker for containerization, facilitating consistent deployment across environments. Knowledge of PuTTY for secure remote connections.
- **Cloud Services**: Experience with Snowflake and Snowpark for cloud-based data warehousing and analytics.